Group Proposal: ScamShield US

Generative AI & Natural Language Processing — Final Project Proposal

1. Problem and motivation

Consumers frequently receive phishing, government/banking impersonation, investment, and identity-theft themed SMS messages. We propose **ScamShield US**, an educational agentic RAG assistant that (1) classifies suspicious messages and (2) answers user questions using official public guidance. We selected this problem because it combines classical NLP classification with modern transformer retrieval and generation, maps to real consumer harm, and fits the course stack constraint of AWS EC2 + Amazon S3 with open-source models only (no Bedrock / paid LLM APIs).

2. Dataset

We use two public SMS corpora: the **UCI SMS Spam Collection** (Category/Text ham–spam) and the **Super SMS Dataset** (SMSes/Labels 0/1). Raw files live under `data/raw/` in S3. We merge, deduplicate, and create train/validation/test splits in `data/processed/`. Separately, we seed ≥20 URL-backed official knowledge documents (FTC, CISA, SSA, IRS, etc.) for RAG under `data/knowledge/`.

3. NLP methods from the course concept list

We will use both classical and pretrained transformer methods:

- **Rule / classical models:** TF-IDF features with Logistic Regression and Linear SVM (Traditional NLP).
- **Pretrained Transformers:** DistilBERT for advanced classification; MiniLM embeddings for dense retrieval; Qwen2.5-1.5B-Instruct as a local decoder LLM for grounded answers.
- **Hybrid cascade:** keep Traditional predictions when confident; escalate weak cases to DistilBERT.
- **Retrieval-augmented generation:** hybrid BM25 + FAISS (top_k=5) over official guidance, then Qwen answers with citations.

We intentionally emphasize classical + transformer pretrained models over RNNs/autoencoders for this product-facing pipeline. Lightweight interpretability is provided via term-frequency dashboards and model comparison tables (full LIME/SHAP analysis is noted as future work).

4. Packages and why

- **scikit-learn / joblib** — classical classifiers and pipelines.
- **transformers / torch / sentence-transformers / faiss / rank-bm25** — DistilBERT, MiniLM, Qwen, hybrid retrieval.
- **FastAPI + Uvicorn + Angular** — API and UI for demos.
- **boto3** — only S3 access path for durable data/models.
- **uv** — reproducible environment via `pyproject.toml` / `uv.lock` / `uv sync`.

5. NLP tasks

- Text classification (binary ham/spam and multi-class scam subtypes).
- Information retrieval (hybrid sparse+dense).
- Grounded answer generation / educational QA (decoder LLM).

6. Evaluation metrics

Primary metric: **macro-F1** (rare scam subtypes matter). We also report accuracy, weighted-F1, precision/recall, and confusion matrices on a held-out test split. Hybrid cascade margin is selected on validation. RAG quality is assessed qualitatively via cited answers in the UI and retrieval provenance.

7. Rough schedule

Week	Milestone
1	Proposal, dataset acquisition, repo skeleton, S3 layout
2	Baseline Traditional classifiers + DistilBERT training
3	Knowledge seeding, hybrid index, Qwen RAG answers
4	API + Angular UI, evaluation tables, dashboard viz
5	Report, presentation, GitHub deliverable folders, polish